

S Ajeth

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Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Engineering

Aug 2023 – July 2027

CGPA:8.61(Up-to 5th Semester)

Experience

Embedded Systems and Hardware Intern, Sony SSUP – Coimbatore, TN

Jan 2025 – Sep 2025

- Contributed to Phase 4 of “*An Edge-based Cyber-Physical System for Smart Polyhouse Solar Drying of Agricultural Food Products*”, focusing on embedded hardware integration, firmware development, and system optimization
- Gained hands-on experience with Sony Spresense, ESP32, RaspberryPi, and various GSM modules for embedded system applications
- Assisted in ensuring long-term operation and monitoring of sensor nodes for continuous food product monitoring without system failure
- Assisted in migrating project architecture, improving scalability and performance
- Developed and deployed CNN models for food product monitoring

Projects

Presense360 – Mobile Edge Based Facial Attendance System | [GitHub-App](#) | [GitHub-Web](#) | [GitHub-Server](#)

- Developed an edge-based facial recognition attendance management system as both a mobile application and a web application.
- Built the mobile app using Flutter, enabling employees to mark attendance with geofencing and real-time face authentication
- Developed the web platform with React.js and TailwindCSS, providing administrators with tools to manage users, and monitor logs
- Implemented backend services with Node.js and Express.js, and used PostgreSQL for data storage
- Tools Used: Flutter, React.js & TailwindCSS, Node.js, Express.js, PostgreSQL

Smart Traffic Light System | [GitHub](#)

- Developed an intelligent traffic light control system using Arduino Mega
- Implemented embedded firmware to process real-time sensor input and control light sequencing of traffic light system
- Optimized signal switching to reduce idle time and improve traffic flow efficiency
- Tools Used: Arduino Mega

Road Management and Traffic Optimization | [GitHub](#)

- Developed a Traffic Management System in Python using data structures such as graphs, hash maps, and linked lists
- Implemented Dijkstra's, Prim's, and Floyd-Warshall algorithms for optimized route planning
- Tools Used: Python, Graphs, Hash Map, Linked List, Dijkstra's, Prim's and Floyd-Warshall Algorithms

SmartExpense – AI-Powered Expense Management System | [GitHub-App](#) | [GitHub-Server](#)

- Vibe coded a cross-device AI-powered expense tracking application with group-based financial management
- Integrated Phi-3 for natural language expense parsing, combining LLM-based classification with deterministic numeric extraction for reliability
- Tools Used: Anti-Gravity

Technical Skills

Areas of Interest: Full Stack Development, Mobile app Development, Embedded Firmware and Hardware Development

Languages: Java, Python, C, JavaScript, HTML & CSS, Dart, SQL

Embedded Technologies: Sony Spresense, Arduino, ESP32, RaspberryPi, STM32, Electrosmith Daisy Seed

Technologies: Node.js, Express.js, Flutter, React.js, Tailwind CSS, PostgreSQL